Derivation of `EpiLink` compatibility score

For a sampled pair (i, j), let t_{ij} denote the testing-time difference and g_{ij} the observed consensus-level genetic distance. `EpiLink` compares a finite set of recent latent transmission scenarios linking the pair, computes a compatibility score for each scenario, and reports the compatibility of a user-specified target scenario.

Observed quantities and timing model

Let $t_{\text{test},i}$ and $t_{\text{test},j}$ denote the testing dates, measured in days on a common time scale, and define

$$t_{ij} = t_{\text{test},j} - t_{\text{test},i}. \tag{1}$$

Let g_{ij} denote the observed consensus-level genetic distance, measured in nucleotide differences.

We implement the variable infectiousness $E/P/I$ model of Hart et al. [2021]. Infection is partitioned into latent (E), presymptomatic infectious (P), and symptomatic infectious (I) stages with Gamma-distributed durations y_E, y_P, and y_I, and we extend this temporal model with a Gamma-distributed testing delay x_{test} from symptom onset to testing. Writing y_{toit} for the time from onset of infectiousness to transmission, the incubation period and generation interval are

$$\tau_{\text{inc}} = y_E + y_P, \tag{2}$$

$$\tau_{\text{gen}} = \tau_{\text{inc}} + y_{\text{toit}}, \tag{3}$$

so that the total time from infection to testing is $\tau_{\text{inc}} + x_{\text{test}}$. The temporal model therefore provides latent random variables such as incubation periods, generation intervals, and testing delays, which are integrated over by Monte Carlo.

Latent transmission scenarios

Let M denote the maximum hidden depth allowed in the latent relationship between cases i and j. The candidate set is

$$\mathcal{S}_M = \{H_{\text{AD}}(m) : m = 0, \ldots, M\} \cup \{H_{\text{CA}}(m_i, m_j) : m_i, m_j \geq 0,\ m_i + m_j \leq M\}. \tag{4}$$

Here, $H_{\text{AD}}(m)$ denotes an ancestor–descendant scenario in which case j descends from case i through m unsampled intermediates, whereas $H_{\text{CA}}(m_i, m_j)$ denotes a common-ancestor scenario in which both sampled cases descend from an unsampled source with branch depths m_i and m_j. The cases $H_{\text{AD}}(0)$ and $H_{\text{CA}}(0, 0)$ correspond to direct transmission and co-primary infection, respectively.

Temporal draws

For a given scenario $s \in \mathcal{S}_M$, let t_x denote the infection time of the latent reference point: the infection time of case i under an ancestor–descendant scenario, or the infection time of the shared unsampled

source under a common-ancestor scenario. Let $A_k(s)$ denote the elapsed time from this reference point to infection of sampled case $k \in \{i, j\}$. Then

$$t_{\text{test},k} = t_x + A_k(s) + \tau_{\text{inc},k} + x_{\text{test},k}, \tag{5}$$

so the model-implied testing-time difference is

$$T_s = A_j(s) - A_i(s) + \tau_{\text{inc},j} + x_{\text{test},j} - \tau_{\text{inc},i} - x_{\text{test},i}. \tag{6}$$

Under $H_{\text{AD}}(m)$,

$$A_i(H_{\text{AD}}(m)) = 0, \tag{7}$$

$$A_j(H_{\text{AD}}(m)) = \sum_{r=0}^{m} \tau_{\text{gen},r}, \tag{8}$$

and therefore

$$T_{ij}(H_{\text{AD}}(m)) = \sum_{r=0}^{m} \tau_{\text{gen},r} + \tau_{\text{inc},j} + x_{\text{test},j} - \tau_{\text{inc},i} - x_{\text{test},i}. \tag{9}$$

Under $H_{\text{CA}}(m_i, m_j)$,

$$A_i(H_{\text{CA}}(m_i, m_j)) = \sum_{r=1}^{m_i+1} \tau^{(i)}_{\text{gen},r}, \tag{10}$$

$$A_j(H_{\text{CA}}(m_i, m_j)) = \sum_{s=1}^{m_j+1} \tau^{(j)}_{\text{gen},s}, \tag{11}$$

so that

$$T_{ij}(H_{\text{CA}}(m_i, m_j)) = \sum_{s=1}^{m_j+1} \tau^{(j)}_{\text{gen},s} - \sum_{r=1}^{m_i+1} \tau^{(i)}_{\text{gen},r} + \tau_{\text{inc},j} + x_{\text{test},j} - \tau_{\text{inc},i} - x_{\text{test},i}. \tag{12}$$

For each scenario s, Monte Carlo simulation yields draws $T_s^{(1)}, \ldots, T_s^{(N)}$ from the induced temporal distribution.

Genetic draws

The genetic component assumes that within-host divergence is negligible on the timescales of interest and that consensus divergence accumulates along transmission-linked lineages. Under scenario s, let B_s denote the effective transmission-related branch length, in days, separating the sampled genomes.

For ancestor–descendant scenarios,

$$B_{H_{\text{AD}}(m)} = \sum_{r=0}^{m} \tau_{\text{gen},r} + \tau_{\text{inc},j} + x_{\text{test},j} - (\tau_{\text{inc},i} + x_{\text{test},i}). \tag{13}$$

For common-ancestor scenarios,

$$B_{H_{\text{CA}}(m_i,m_j)} = \sum_{r=1}^{m_i+1} \tau^{(i)}_{\text{gen},r} + \sum_{s=1}^{m_j+1} \tau^{(j)}_{\text{gen},s} + \tau_{\text{inc},i} + x_{\text{test},i} + \tau_{\text{inc},j} + x_{\text{test},j}. \tag{14}$$

For each scenario s, Monte Carlo simulation yields draws $B_s^{(1)}, \ldots, B_s^{(N)}$.

Let r denote the median substitution rate per site per year and L the genome length in sites. The corresponding per-genome daily substitution rate is

$$\lambda = \frac{rL}{365}. \tag{15}$$

Under a relaxed uncorrelated log-normal clock (UCLN), one instead samples

$$r^{(n)} \sim \text{LogNormal}(\mu_r, \sigma_r^2), \tag{16}$$

with μ_r chosen so that the median equals the specified substitution rate r, and sets

$$\lambda^{(n)} = \frac{r^{(n)} L}{365}. \tag{17}$$

Given branch-length draw $B_s^{(n)}$ and clock-rate draw $\lambda^{(n)}$, the genetic draw used for scoring is

$$G_s^{(n)} = \lambda^{(n)} B_s^{(n)} \tag{18}$$

under a deterministic mutational process, and

$$G_s^{(n)} \mid B_s^{(n)}, \lambda^{(n)} \sim \text{Poisson}\Big(\lambda^{(n)} B_s^{(n)}\Big) \tag{19}$$

under a stochastic mutational process.

Compatibility scores

For any observed quantity x_{obs} and scenario s with Monte Carlo draws $x_1^{(s)}, \ldots, x_{n_s}^{(s)}$, define the percentile score

$$p_s(x_{\text{obs}}) = \frac{\#\left\{i : x_i^{(s)} \le x_{\text{obs}}\right\}}{n_s}, \tag{20}$$

and the corresponding compatibility score

$$C_s(x_{\text{obs}}) = 1 - 2\left|p_s(x_{\text{obs}}) - 0.5\right|. \tag{21}$$

Applying this definition to the temporal draws gives

$$p_{T,s}(i,j) = \frac{\#\left\{n : T_s^{(n)} \le t_{ij}\right\}}{N}, \qquad C_{T,s}(i,j) = 1 - 2\left|p_{T,s}(i,j) - 0.5\right|, \tag{22}$$

and applying it to the genetic draws gives

$$p_{G,s}(i,j) = \frac{\#\left\{n : G_s^{(n)} \le g_{ij}\right\}}{N}, \qquad C_{G,s}(i,j) = 1 - 2\left|p_{G,s}(i,j) - 0.5\right|. \tag{23}$$

The overall compatibility assigned to scenario s is the product of the temporal and genetic compatibilities,

$$C_s(i,j) = C_{T,s}(i,j)\, C_{G,s}(i,j). \tag{24}$$

Target subset and computation

Let $\mathcal{S}_\star \subseteq \mathcal{S}_M$ denote a user-specified target subset chosen when the `EpiLink` object is constructed. The reported target score is

$$\text{score}_{\mathcal{S}_\star}(i,j) = \sum_{s \in \mathcal{S}_\star} C_s(i,j). \tag{25}$$

When $\mathcal{S}_\star$ contains a single element, this reduces to the original single-scenario score. The implementation also returns $C_s(i,j)$ for every $s \in \mathcal{S}_M$.

For each scenario $s \in \mathcal{S}_M$, `EpiLink` precomputes and stores N draws of T_s and B_s at construction time. It then converts the cached branch-length draws into cached mutation draws $G_s^{(n)}$ using the chosen mutational process. Each pairwise evaluation therefore only requires percentile calculations against the cached draws, which is well suited to large datasets.

References

William S Hart, Philip K Maini, and Robin N Thompson. High infectiousness immediately before covid-19 symptom onset highlights the importance of continued contact tracing. *eLife*, 10, April 2021. ISSN 2050-084X. doi: 10.7554/elife.65534. URL `http://dx.doi.org/10.7554/eLife.65534`.